

Jack White

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Education

Purdue University

West Lafayette, IN

B.S. Computer Science and B.S. Artificial Intelligence

2023–2027

- GPA: 3.6/4.0
- Relevant Coursework: Introduction to Robotics, Imitation Learning for Robotics (current), Operating Systems, Systems Programming, Analysis of Algorithms, Data Mining and Machine Learning, Probability

Professional Experience

BPM Microsystems

Houston, TX

Automation / AI Software Engineering Intern

Jul–Aug 2025; May–Aug 2026

- Built **AlgoBuilder**, an autonomous engineering workflow for developing and validating semiconductor programming algorithms, integrating Cursor agents, Salesforce through Model Context Protocol (MCP), BPWin REST/WebSocket APIs, automated flash QA, persistent job state, Google Chat hardware hand-offs, and code-review submission.
- Built **PDR Builder**, a datasheet-to-design automation pipeline that extracts semiconductor pin, protocol, and electrical requirements, applies deterministic validation, routes uncertain results through human review, and emits production .pdr and Excel engineering artifacts.
- Validated PDR automation on a historical corpus: **27/27 Tier-C designs passed with 0.09 noise errors/design**; model-backed pin-extraction evaluations recorded zero silent errors on scored cases.
- Responsible for the **BPWin REST API 1.2–1.9** upgrade supporting unattended hardware operation, including job observability, remote dialog control, device/socket discovery, NAND ranges, algorithm-database reindexing, and serialization capability validation; authored/used a **65-case** feature test plan across automated and hardware lanes.
- Previously modernized BPM's internal AI assistant, **Chip**, around MCP; developed system/unit tests, integrated Salesforce and Google Drive tools, secured API credentials, and reduced model-call costs.

Research Experience

Purdue University – Prof. Yung-Hsiang Lu

West Lafayette, IN

Computer Vision Research

Aug 2025–Dec 2025

- Implemented computer-vision methods for automated piano-performance transcription to MIDI, supporting evaluation of generative systems that synthesize musical-performance video from sheet music.

Selected Projects

ControVirtual – XR Robotic Teleoperation and Robot Learning

Apr 2026

- Built a robotics integration stack connecting a Meta Quest Unity client, spoken commands, a Python WebSocket bridge, live OpenCV camera streaming, robot telemetry, and a browser-based digital twin.
- Fine-tuned the **SmolVLA vision-language-action model** on **50 teleoperated episodes** of an SO-101 robot-arm manipulation task and built inference scaffolding combining language instructions, visual observations, and robot state.
- Implemented runnable voice-command, camera, digital-twin, and simulated telemetry subsystems; full policy-to-physical-robot action dispatch remained experimental.

TradingSim – Historical Market Microstructure Replay Simulator

2026

- Built a browser-based replay simulator using Databento Level-2 depth, trade tape, OHLCV data, and optional news, with strict timestamp ordering and no-lookahead guarantees.
- Implemented latency-aware, depth-limited fills, passive-order queue modeling, commissions/fees, risk gates, atomic data writes, and **126 automated tests** across Python and frontend code.

RCBot – Event-Driven Autonomous Trading System

2026

- Built a standalone small-cap momentum bot with a shared event contract for historical replay and an Interactive Brokers Gateway path, incremental feature calculation, setup detectors, risk governor, position management, and simulated execution.
- Added experiment tooling for threshold sweeps, labelled candidate analysis, execution stress tests, and a run ledger storing fully resolved configurations and results for reproducible comparisons.

FilingEdge – Financial Machine Learning Research System

2026

- Built a point-in-time ML pipeline predicting post-filing abnormal returns using SEC filings/XBRL fundamentals, macroeconomic series, and market data, stored in DuckDB.
- Implemented purged walk-forward cross-validation, leakage assertions, logistic-regression and LightGBM models, transaction-cost-aware backtesting, and a Streamlit research dashboard.

Recipe Finder – Full-Stack Cookbook Search and Matching Platform

2026

- Built a FastAPI/PostgreSQL/Redis backend and React/TypeScript frontend that ingests PDF, EPUB, and TXT cookbooks, extracts structured recipes, and matches them against pantry ingredients.
- Implemented asynchronous ingestion jobs, versioned REST APIs, JWT authentication with rotating HttpOnly refresh cookies, Docker deployment, and private-by-default data services.

SlopMop – AI-Content Detection Browser Extension

2026

- Built a browser-extension/client-server system that flags likely AI-generated text and images in social feeds, with asynchronous backend inference, source lookup, caching, usage history, and incognito-session behavior.
- Developed a custom PyTorch/Hugging Face text-classification model and integrated text/image detection into a FastAPI backend deployed on AWS.

C Shell

Oct–Dec 2024

- Implemented a Unix-style shell in C using Lex and Yacc, recreating core and advanced Bash-like command parsing and execution behavior from scratch.

Honors & Additional Experience

- **Honorable Mention – Best Use of AMD Tech**, StarkHacks 2026, Purdue University, for ControVirtual.
- **Music Teacher, Redwood High School** (Apr–Jun 2023): taught music theory, ear training, and piano technique in classical, rock, and jazz contexts.
- **Independent Day Trading** (Jun 2022–Present): developed and refined momentum-trading workflows using high-of-day scanning and custom technical indicators.

Technical Skills

Programming: Python, C, C++, C#, Java, TypeScript/JavaScript, SQL, Assembly

AI / Machine Learning: PyTorch, Hugging Face, LightGBM, LeRobot, SmolVLA, LLM APIs, Model Context Protocol (MCP), pandas, NumPy

Robotics / Computer Vision: OpenCV, Unity, Meta XR, vision-language-action models, robot telemetry, digital twins

Systems / Web: Linux, REST APIs, WebSockets, FastAPI, Flask, React, Docker, PostgreSQL, Redis, DuckDB, AWS, Git, Mercurial

Hardware / Semiconductor: semiconductor device programming, NOR/NAND flash, microcontrollers, programmer hardware

Languages: English (fluent), Spanish (proficient)